**Biochemistry Cume**

**Total Points**: 100 points

1. **A**. Consider the following dissociation of an ionizable group, HA,

$$HA \overset{K_a}{\leftrightarrow} H^+ + A^-,$$

Derive an equation to calculate the fraction of HA and $A^-$ formed at any pH (**25 points**)

**B.** Calculate the net charge on the dipeptide, K-E, at pH 4.2. (Given the following pKa values of α-carboxyl, α-amino, β-carboxyl, γ-carboxyl, thiol, phenolic, ε-amino, imidazole, and guanido groups of amino acids in peptides are, 2.3, 8.6, 3.2, 4.2, 8.0, 10.5, 10.1, 6.0, 11.2, respectively). (**25 points**) [***To receive credit, should show the relevant work that was used to arrive at the solution to this problem***]

**C.** Predict the direction of mobility of dipeptide, at pH 4.2, when it is subjected to paper electrophoresis (**5 points**)

2. **A**. Calculate the pH at which the net charge on glutamic acid will be zero. (Given the following pKa values of α-carboxyl, α-amino, β-carboxyl, γ-carboxyl, thiol, phenolic, ε-amino, imidazole, and guanido groups of amino acids in peptides are, 2.3, 8.6, 3.2, 4.2, 8.0, 10.5, 10.1, 6.0, 11.2, respectively) (**30 points) [*To receive credit, the net charge on each ionic species that are formed during the course of titration of glutamic acid with the base should be calculated*].**

**B.** Draw a cartoon depicting the titration curve of glutamic acid with a base. Should show the percentage population of undissociated and dissociated species formed during the course of titration **(15 points)**